

# Network Tariff Application Guide

For the regulatory year 1 July 2012 to 30 June 2013

**As Approved by the AER**

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**AURORA ENERGY PTY LTD****Network Tariff Application Guide**

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## **1. Introduction**

This 2012-13 Network Tariff Application Guide outlines Aurora's network tariff terms and conditions for Standard Control Services and applies from 1 July 2012 to 30 June 2013.

This 2012-13 Network Tariff Application Guide provides the network use of system (NUoS) tariffs for distribution use of system (DUoS) and transmission use of system (TUoS) charges applied by Aurora for all distribution connected customer sites.

This 2012-13 Network Tariff Application Guide provides information on how Aurora assigns customers to tariff classes and the review process which is undertaken where a customer objection is received in writing.

Further information on Aurora's network tariffs can be found at Aurora's website at [http://www.auroraenergy.com.au/electricity\\_network](http://www.auroraenergy.com.au/electricity_network) and in Aurora's 2012-13 Pricing Proposal.<sup>1</sup>

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<sup>1</sup> Aurora Energy Pricing Proposal, 1 July 2012 – 30 June 2013, May 2012.

## **2. Application of network tariffs**

### **2.1. Aurora**

All references to Aurora within this Network Tariff Application Guide, unless otherwise stated, are to Aurora in its capacity as a licensed distribution network service provider in the Tasmanian jurisdiction only.

### **2.2. Goods and service tax (GST)**

The NUoS prices and regulated network tariffs published by Aurora, unless otherwise stated, are exclusive of GST.

### **2.3. Time zones**

All times reflected in this document are in Australian Eastern Standard Time (AEST) unless otherwise specified.

### **2.4. Metering services**

The standard charge for the provision of metering services may apply, depending upon the contestability status of the customer.

Where a customer has entered into a market contract in accordance with the Contestable Customer Regulations the standard metering services charge may not apply. Charges for the provision of metering services will apply should Aurora be appointed as the metering provider (MP) and will be negotiated in accordance with the MP contract.

In all other cases the standard metering services charge will apply.

Further information on Aurora's metering services tariffs can be found in Aurora's 2012-13 Metering Services Application and Price Guide.

### **2.5. Choice of network charges**

It is a matter for the customer's retailer of choice to determine the network tariff that is appropriate for their needs. Transfers between network tariffs will apply from the beginning of the normal billing period following the month in which written advice is received by Aurora and transfers will not have any retrospective effect.

Aurora reserves the right to review a customer's network tariff in the event of any electrical load changes and will notify the customer's retailer of choice regarding any resulting network tariff pricing changes. The final decision on the appropriate network tariff shall be at the discretion of Aurora.

There will be instances where a customer may have a separate connection agreement with Aurora and in these instances Aurora may directly invoice the customer for network use and the customer's retailer of choice will only provide energy related commercial services.

### **2.6. Obsolete tariffs**

There are a number of network tariffs that Aurora has classified as obsolete. An obsolete network tariff is no longer available to any new customer. Existing customers may remain on an obsolete tariff providing no alteration is made to the customer's installation.

Customers (and/or the customer's retailer) that choose to transfer from an obsolete network tariff will lose all rights to all obsolete network tariffs at that installation, i.e. the entire installation will be required to move to currently available published network tariffs.

### **2.7. Mid month change of retailer of choice**

When a customer elects to change their retailer of choice at any time during a calendar month, NUoS invoices will be allocated pro-rata between the retailers involved.

NUoS invoices will use the "actual readings" obtained on the day of transfer.

Unmetered electricity supplies will be transferred on a pro rata basis based on the number of days in the month the customer was contracted with each retailer. It will not be possible to transfer unmetered electricity supplies between retailers on the same business day.

### **2.8. Standby electricity supply**

Where customers with critical electricity supply needs have a requirement for standby electricity supply capability, applicable network charges will be negotiated on the basis of the assets involved and the amount of feeder capacity required to be kept in reserve to accommodate the standby supply.

### **2.9. Embedded generation**

NUoS charges for Embedded Generation will be individually calculated (refer to section 30 of this Guide).

### **2.10. Published network tariffs**

Aurora's current published network tariff prices can be found on Aurora's website at [http://www.auroraenergy.com.au/electricity\\_network](http://www.auroraenergy.com.au/electricity_network).

### **3. Assigning and reassigning customers to tariff classes**

Aurora assigns customers to tariffs on the basis of usage and size. Customers are assigned into one of following tariff classes:

- ITC;
- HV;
- irrigation;
- large LV;
- small LV;
- residential;
- uncontrolled energy;
- controlled energy;
- unmetered;
- streetlight; and
- embedded generation.

Customers are assigned to at least one tariff class. Assignment to tariff classes is based on:

- the nature of the customer's connection;
- the customer's forecast usage and size; and
- the premise that the same connection and usage profiles are treated on a consistent basis.

#### **3.1. Reassignment of network tariffs**

Customers seeking tariff reassignment must:

- (a) be eligible for tariff reassignment; and
- (b) provide Aurora with one month's written notification; and
- (c) pay any applicable tariff alteration fee<sup>2</sup>.

Customers must remain on the reassigned network tariff for a minimum of 12 months unless otherwise agreed with Aurora. This condition prevents customers from changing network tariffs to take advantage of seasonal variations in prices according to their individual load; thereby bypassing payment that reflects use of the distribution network over a full 12 month cycle.

A tariff reassignment request may be made:

- (a) through the customer's retailer; where:
  - (i) the retailer will notify Aurora; or

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<sup>2</sup> Aurora's fee-based services tariffs for tariff alterations are discussed in Aurora's Fee-based Services Application and Price Guide.

- (b) through Aurora, where:
  - (i) Aurora will advise the customer's retailer.

Exceptions to the above conditions will only be made at Aurora's discretion where it can be demonstrated that to not do so would result in unreasonable penalties or impose hardship on the customer.

## **4. Maximum demand application**

For the purposes of determining a customer's maximum demand for network tariff application the following rules will apply.

### **4.1. Definition of maximum demand**

Maximum demand means the electrical demand measured in kiloVoltAmps (kVA) or kiloWatts (kW) and is calculated as the energy consumption recorded over the demand integration period divided by the demand integration period in hours. Aurora's demand integration period is 15 minutes.

The value is used for setting the maximum demand charges to be paid by the customer during each billing period.

### **4.2. Calculation of maximum demand**

Where maximum demand is used as the basis for network tariff charges, the calculation is determined by using the any-time maximum demand (ATMD) for the installation during the day.

Where the maximum demand charge is for the entire billing period, the ATMD of an installation during the billing period is taken to be the largest value of the electrical demand for that installation during that billing period.

Where the maximum demand charge is for each day during the billing period, the ATMD of an installation during the day is taken to be the largest value of the electrical demand for that installation during that day of the billing period.

### **4.3. Increases in electrical demand**

Where a customer requests a change in network tariff due to an increase in electrical demand at their connection point, the customer must provide 20 business days written advice (prior to the commencement of the next billing period) to Aurora detailing their new requirement. Aurora will notify the customer in writing within 10 business days advising any revised charges or tariff reassignment.

The increased level of electrical demand shall apply from the commencement of the next billing period, subject to any required works being completed by Aurora.

### **4.4. Temporary increases in maximum demand**

Temporary increases in electrical demand will:

- be subject to negotiation and approval by Aurora;
- be defined in terms of "additional demand" for a specific period and charged at an agreed demand charge rate;
- apply for one full billing period, except in the case of commissioning of new plant, in which case the duration of the temporary increase may be extended for the duration of the commissioning period; and
- be limited to one occurrence each 12 months or as otherwise agreed with Aurora.

#### **4.5. Reduction in maximum demand**

Where a customer requests a change in network tariff due to a reduction in electrical demand at their connection point, the customer must provide six months written advice (prior to the commencement of the next billing period) to Aurora detailing their new requirement. Aurora will notify the customer in writing within 60 days advising any revised charges or tariff reassignment.

The decreased level of electrical demand shall apply from the commencement of the billing period advised by Aurora as part of the notification of Aurora's acceptance of the reduced demand.

However, following the installation by a customer of load management equipment approved by Aurora or the implementation of a demand management initiative approved by Aurora, the six month notice period referred to above may be reduced at the discretion of Aurora.

## **5. System of assessment and review of the basis on which a customer is charged**

Aurora's process for setting a customer's specified demand, confirming a customer's specified demand at the start of each regulatory year and assessing a customer's request for change in specified demand during the regulatory year is outlined below.

### **5.1. Setting a customers specified demand**

Customers on certain network tariffs are able to agree, or nominate, with Aurora a specified demand for their electrical installation. Once agreed this specified demand is used in the calculation of demand charges for the customer.

Specified demands for all new customers are established as a component of the customer connection process and remain in force until the customer requests a change in that specified demand.

Aurora will review each existing customer's specified demand annually coinciding with the preparation of Aurora's pricing proposal. This assessment is based on historical data and tariff specifications for each customer on their specified demand related network tariff.

### **5.2. Confirming a customer's specified demand at the start of each regulatory year**

Confirmation of a customer's specified demand is completed through a letter written to each customer (and the customer's retailer) during March each year. This letter seeks the customer's confirmation that the customer wishes to amend their current specified demand. Aurora requires that each customer should respond within 10 business days, or the current specified demand will continue to apply.

The customer letter confirms:

- the network tariff the customer has been assigned or reassigned; and
- that the specified demand will apply for the 12 months from 1 July that year.

A further confirmation letter is sent to the customer (and the customer's retailer) detailing the nominated specified demand and prices that will apply following the AER's approval of Aurora's pricing proposal.

Particular customer specific demands are kept confidential

### **5.3. Assessing a customer's request for change in specified demand during the year**

Aurora will assess a customer's request in line with section 3 of this document.

## **6. Procedure for reviewing complaints and disputes**

Aurora will ensure that all complaints and disputes are dealt with in accordance with its standard complaints and dispute resolution policy and procedures. Aurora's dispute resolution policy is reviewed annually and published on Aurora's website.

### **6.1. Internal procedure for reviewing objections**

Where Aurora receives written notification that a customer has an objection to a proposed tariff assignment or reassignment, the following additional procedures will be followed.

An initial review process will be performed by the customer's retailer and forwarded to Aurora for review. This initial review by the retailer should include the proposed tariff assignment and an indication of the customer's anticipated annual consumption and any time maximum demand for the installation.

Aurora will then undertake the following internal review process:

- (a) the customer's written objection will be reviewed by Aurora and the review will be completed within 10 business days of receipt of the customer's written objection;
- (b) additional information provided by the customer (and/or the customer's retailer) will be considered;
- (c) Aurora will determine the energy and/or demand usage for the customer based on:
  - (i) customer (and/or retailer) information; or
  - (ii) Aurora's historical or estimated energy consumption data;
- (d) an assessment of the customer's connection to the network will be made;
- (e) Aurora will determine the tariff assignment that should apply;
- (f) the proposed tariff assignment will be reviewed and approved by the Manager Revenue Assurance; and
- (g) the customer (and/or customer's retailer) will be notified in writing of the tariff assignment review outcomes within five business days of completion of the Aurora review.

### **6.2. Objection not resolved to satisfaction of customer under internal review process**

If the customer objection to tariff assignment or reassignment to a specific tariff class, after applying Aurora's internal review process as detailed above, is not resolved to the customer's satisfaction, the customer is entitled to seek resolution through the following:

- if the resolution of the dispute is within the jurisdiction of the Energy Ombudsman Tasmania, the customer is entitled to escalate the matter to the Energy Ombudsman Tasmania; or

- if the objection is not resolved to the satisfaction of the customer under Aurora's internal review system and/or the Energy Ombudsman Tasmania, then the customer is entitled to seek a decision of the AER via the dispute resolution process available under Part 10 of the National Electricity Law.

### **6.3. Final tariff class assignment**

#### **6.3.1. Initial tariff assignment**

Where a customer has initially been assigned to a network tariff as a component of that customer's connection to the distribution network, that tariff assignment will remain in force until the resolution of any objection to that tariff assignment, in accordance with these procedures.

Should the resolution of the customer's objection result in a change in network tariff assignment that tariff assignment will be back-dated to the original date of assignment and the customer's account will be adjusted in the next billing period.

#### **6.3.2. Tariff reassignment**

Where a customer has been notified that they will be reassigned to a network tariff and the customer objects to that tariff assignment, that tariff reassignment will not occur until the resolution of any objection to that tariff reassignment, in accordance with these procedures.

Should the resolution of the customer's objection result in a tariff reassignment, that tariff reassignment will occur at the commencement of the next billing period for the customer or the originally notified date; whichever is the later.

## 7. Network tariffs for Standard Control Services

Table 1 shows the Standard Control Services network tariffs that Aurora will offer for the 2012-13 regulatory year.

**Table 1: Standard Control Services network tariffs**

| Description                               | Aurora Code  | Type                         |
|-------------------------------------------|--------------|------------------------------|
| General Network - Residential             | N01          | Published Tariff             |
| General Network - Business                | N02          | Published Tariff             |
| General Network – Business, Nursing Homes | N02a         | Published<br>Obsolete Tariff |
| General Network – Business, Curtilage     | N02b         | Published<br>Obsolete Tariff |
| LV kW Demand                              | N03          | Published<br>Obsolete Tariff |
| Uncontrolled Energy                       | N05          | Published Tariff             |
| Controlled Energy                         | N06          | Published Tariff             |
| Controlled Energy                         | N06a         | Published Tariff             |
| Small LV Unmetered                        | N07          | Published Tariff             |
| LV Day/Night (Irrigation)                 | N08          | Published<br>Obsolete Tariff |
| LV Irrigation (ToU)                       | N08a<br>N08b | Published Tariff             |
| LV kVA Demand                             | N09          | Published Tariff             |
| HV kVA Demand                             | N10          | Published<br>Obsolete Tariff |
| HV kVA Specified Demand                   | N10s         | Published Tariff             |
| HV kW Demand                              | N11          | Published<br>Obsolete Tariff |
| LV PAYG                                   | N13          | Published<br>Obsolete Tariff |
| LV ToU - Business                         | N13b<br>N13c | Published Tariff             |
| LV ToU - Residential                      | N13r         | Published Tariff             |
| HV kVA Specified Demand (>2.0 MVA)        | N15          | Published Tariff             |
| Street Lighting                           | N20          | Published Tariff             |
| Import Energy                             | N21          | Published Tariff             |
| Individual network tariff Calculation     | ITC          | Negotiated Tariff.           |

## **8. General Network – Residential (N01)**

This network tariff is for low voltage installations that are premises used wholly or principally as Private Residential Dwellings.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

Farm outbuildings may be connected on this network tariff provided that the connection is through the meters of the farm residence.

This network tariff may also be used in conjunction with the following additional network tariffs:

- N05 – Uncontrolled Energy;
- N06 – Controlled Energy; and
- N06a – Controlled Energy.

A Type 6 meter is the minimum required for installations on this network tariff.

## **9. General Network - Business (N02)**

This network tariff is for low voltage installations that are not Private Residential Dwellings.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

This network tariff may also be used in conjunction with the following additional network tariffs:

- N05 – Uncontrolled Energy;
- N06 – Controlled Energy; and
- N06a – Controlled Energy.

A Type 6 meter is the minimum required for installations on this network tariff.

## **10. General Network – Business, Nursing Homes (N02a)**

This network tariff is obsolete and therefore no longer available to new customers.

This network tariff is for low voltage installations that are registered as aged care facilities.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

This network tariff may also be used in conjunction with the following additional network tariffs:

- N05 – Uncontrolled Energy; and
- N06 – Controlled Energy.

A Type 6 meter is the minimum required for installations on this network tariff.

The step energy charges within this network tariff will be increased each regulatory year by an amount equal to the change in CPI plus an additional 6 per cent until such time as those step energy charges achieve parity with the energy charge within network tariff N02 – General Network Business. When this network tariff achieves parity with the network tariff N02 – General Network Business, this network tariff will be discontinued and will not be available to any customer. Aurora will write to any customers remaining on this network tariff, at this time, advising that Aurora intends to reassign those customers to network tariff N02 – General Network Business.

## **11. General Network – Business, Curtilage (N02b)**

This network tariff is obsolete and therefore no longer available to new customers.

This network tariff is for low voltage rural installations having a single connection point but requiring more than one meter due to site layout.

The single connection point must supply an installation qualifying for, and being supplied network tariff, N01 – General Network Residential.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

With the exception of the preceding clarification, this network tariff may not be used in conjunction with any other network tariff.

A Type 6 meter is the minimum required for installations on this network tariff.

The daily charge within this network tariff will be increased each regulatory year by an amount equal to a decreasing annual 10 per cent discount of the network tariff N02 – General Network Business daily charge until such time as the daily charge achieves parity with the daily charge within network tariff N02 – General Network Business. When this network tariff achieves parity with the network tariff N02 – General Network Business, this network tariff will be discontinued and will not be available to any customer. Aurora will write to any customers remaining on this network tariff, at this time, advising that Aurora intends to reassign those customers to network tariff N02 – General Network Business.

## **12. Uncontrolled Energy (N05)**

This network tariff is for low voltage installations.

### **12.1. General conditions**

#### **12.1.1. Private Residential Dwellings**

In installations that are Private Residential Dwellings, this network tariff:

- is for water heating and/or residential space heating and/or domestic indoor pool heating only; and
- may only be used if the installation also has a current connection on network tariff N01 – General Network Residential.

#### **12.1.2. Other installations**

In installations that are not Private Residential Dwellings, this network tariff:

- is for water heating only; and
- may only be used if the installation also has a current connection on network tariff N02 – General Network Business or N02a – General Network Business, Nursing Homes.

#### **12.1.3. All installations**

With the exception of thermal-storage space heaters or thermal-storage water heaters, this network tariff may not be applied to any apparatus also connected under another network tariff.

This network tariff is not available in its own right and must be used in conjunction with one of the following additional network tariffs:

- N01 – General Network Residential;
- N02 – General Network Business; and
- N02a – General Network Business, Nursing Homes.

A Type 6 meter is the minimum required for installations on this network tariff.

### **12.2. Requirements of water heating systems**

#### **12.2.1. Private Residential Dwellings**

In installations that are Private Residential Dwellings, for connection on this network tariff, the water heating systems:

- must comply with AS 1056, Storage water heaters; and
- should comply with AS/NZS 3500.4:2003, Plumbing and drainage – Heated waters services and AS 3500.4.1 – 1997, National Plumbing and Drainage – Hot water supply systems – Performance requirements; and
- must, if the delivery rating of the water heating system is less than or equal to 500 litres, have an electric heating unit rating not exceeding 16 Watts per litre; or
- must, if the delivery rating of the water heating system is greater than 500 litres, have an electric heating unit rating not exceeding 32 Watts per litre.

Non-compliant systems may be refused connection or be disconnected.

Where a Private Residential Dwelling has a water storage heater installed, and the delivery rating is greater than 20 litres but less than 100 litres, the limit of 16 Watts per litre may be exceeded by that individual water storage heater. Only one water storage unit, with a delivery rating between 20 and 100 litres that exceeds the 16 Watts per litre threshold may be installed at a Private Residential Dwelling.

#### **12.2.2. Other installations**

In installations that are not Private Residential Dwellings, for connection on this network tariff, the water heating systems:

- must comply with Australian Standard 1056, Storage Water Heaters; and
- should comply with AS/NZS 3500.4:2003, Plumbing and drainage – Heated waters services and AS 3500.4.1 – 1997, National Plumbing and Drainage – Hot water supply systems – Performance requirements, and
- must, if the delivery rating of the water heating system is less than or equal to 500 litres, have an electric heating unit rating not exceeding 16 Watts per litre; or
- must, if the delivery rating of the water heating system is greater than 500 litres, have an electric heating unit rating not exceeding 32 Watts per litre.

Non-compliant systems may be refused connection or be disconnected.

Where an installation that is not a Private Residential Dwelling has two or more water storage heaters installed, and the combined delivery rating is greater than 500 litres, the limit of 32 Watts per litre may be exceeded by an individual water storage heater provided that the ratio of the total wattage of all the water heating units to the total delivery rating does not exceed 32 Watts per litre.

#### **12.3. Dairy water heaters**

Dairy water heaters containing main and booster heating units may have both heating units connected under this network tariff.

Dairy water heaters are not required to comply with the AS 1056.

The electric heating unit ratings detailed in section 12.2.2 do not apply to dairy water heaters.

#### **12.4. Requirements of residential space heating systems**

Permanently installed “wired-in” electric heater(s) may be eligible for this network tariff on condition that the wiring of any electric heater is installed by a registered electrician in accordance with AS/NZS 3000 wiring rules and associated regulations and acts, and one of the following conditions are met:

- if a residence has a permanently installed “wired-in” electric heater with an output of 3.5 kW in a living area, on a single functional switch then this, and any additional permanently “wired-in” space heaters throughout the residence, may be installed on this network tariff; or

- a total rating of at least 5 kW of the same heating system installed throughout the residence. This heating system must be the priority heating system of the main living area and must have a single functional switch in each heated area throughout the residence. However where a ducted heating system is installed, the control switch must be located near the heating unit in order to qualify for this network tariff; or
- heating in secondary areas such as bedrooms and hallways if the residence has off peak storage heating in the living area(s) as its priority source of heating. The secondary heating system should be a permanently connected single propriety heating system with a total of 5 kW or more heating capacity.

### **12.5. Requirements of domestic indoor pool heating systems**

Private domestic indoor swimming pools are allowed under this network tariff if an installation:

- complies with the residential space heating system rules as provided above; and
- has an electrical input power limit of 400 Watt/m<sup>2</sup> of surface area.

### **12.6. Domestic spa systems**

Spas are not eligible for connection to this network tariff.

## **13. Controlled Energy (N06)**

This network tariff is for low voltage installations.

### **13.1. General conditions**

#### **13.1.1. Private Residential Dwellings**

In installations that are Private Residential Dwellings, this network tariff:

- is for water heating and/or residential space heating and/or other “wired in” appliances as approved by Aurora; or
- may be used for heating swimming pools, including those that incorporate a spa. Note that an individual spa from which the water goes to waste after use may not be connected on this network tariff; and
- may only be used if the premises also has a current connection on network tariff N01 – General Network Residential.

#### **13.1.2. Other installations**

In installations that are not Private Residential Dwellings, this network tariff:

- is for water heating and/or space heating and/or other “wired in” appliances as approved by Aurora; and
- may only be used if the premises also has a current connection on network tariff N02 – General Network Business, or N02a – General Network Business, Nursing Homes.

#### **13.1.3. All installations**

With the exception of thermal-storage space heaters or thermal-storage water heaters, this network tariff may not be applied to any apparatus also connected under another network tariff.

In all installations, this network tariff may not be used for circuits supplying general-purpose outlets, other than existing outlets supplied on this tariff.

This network tariff is not available in its own right and must be used in conjunction with one of the following additional network tariffs:

- N01 – General Network Residential;
- N02 – General Network Business; and
- N02a – General Network Business, Nursing Homes.

A Type 6 meter is the minimum required for installations on this network tariff and must have the ability to control energy flows.

### **13.2. Time of use availability**

This network tariff is a “time of use” tariff. Energy to installations connected on this network tariff will be available daily for:

- at least nine hours between 20:00 hours and 07:00 hours the following day; and
- a further two hours between 13:00 hours and 16:30 hours.

Aurora will choose the actual times during the periods that the energy will be available.

### **13.3. Requirements of water heating systems**

In all installations, for connection on this network tariff, the water heating systems:

- must comply with AS 1056, Storage water heaters; and
- should comply with AS/NZS 3500.4:2003, Plumbing and drainage – Heated waters services and AS 3500.4.1 – 1997, National Plumbing and Drainage – Hot water supply systems – Performance requirements.

Non-compliant systems may be refused connection or disconnected.

### **13.4. Requirements of space heating systems**

Permanently installed “wired-in” electric heater(s) may be eligible for this network tariff on condition that the wiring of any electric heater is installed by a registered electrician in accordance with AS/NZS 3000 wiring rules and associated regulations and acts.

### **13.5. Requirements of “wired in” appliances**

Permanently installed “wired-in” appliances may be eligible for this network tariff on condition that the wiring of any appliance is installed by a registered electrician in accordance with AS/NZS 3000 wiring rules and associated regulations and acts.

## **14. Controlled Energy (N06a)**

This network tariff is for low voltage installations.

### **14.1. General conditions**

#### **14.1.1. Private Residential Dwellings**

In installations that are Private Residential Dwellings, this network tariff:

- is for water heating and/or residential space heating and/or other circuits as approved by Aurora;
- may be used for heating swimming pools, including those that incorporate a spa. Note that an individual spa from which the water goes to waste after use may not be connected on this network tariff; and
- may only be used if the installation also has a current connection on network tariff N01 – General Network Residential or N13r – LV ToU Residential.

#### **14.1.2. Other installations**

In installations that are not Private Residential Dwellings, this network tariff:

- is for water heating and/or space heating and/or other circuits as approved by Aurora; and
- may only be used if the installation also has a current connection on network tariff N02 – General Network Business or N13b/N13c – LV ToU Business.

#### **14.1.3. All installations**

In all installations, this network tariff may be used for circuits supplying general-purpose outlets.

This network tariff is not available in its own right and must be used in conjunction with one of the following additional network tariffs:

- N01 – General Network Residential;
- N13r – LV ToU Residential;
- N02 – General Network Business; and
- N13b/N13c – LV ToU Business.

A Type 6 meter is the minimum required for installations on this network tariff, must be capable of recording time of use data and must have the ability to control energy flows.

### **14.2. Time of use availability**

This network tariff is a “time of use” tariff. Energy to installations connected on this network tariff will only be available between 22:00 hours and 07:00 hours the following day.

### **14.3. Requirements of water heating systems**

In all installations, for connection on this network tariff, the water heating systems:

- must comply with AS 1056, Storage water heaters; and

- should comply with AS/NZS 3500.4:2003, Plumbing and drainage – Heated waters services and AS 3500.4.1 – 1997, National Plumbing and Drainage – Hot water supply systems – Performance requirements.

Non-compliant systems may be refused connection or disconnected.

## **15. LV PAYG (N13)**

This network tariff is obsolete and therefore no longer available to new customers.

This network tariff is for low voltage installations that are premises used wholly or principally as Private Residential Dwellings and supplied in accordance with the Aurora Energy Business Pay As You Go (PAYG) product.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

This network tariff may not be used in conjunction with any other network tariff.

Standard metering services do not apply for this tariff and all metering services will be provided by the Aurora Distribution Business.

### **15.1. Requirements of water heating systems**

In all installations, for connection on this network tariff, the water heating systems:

- must comply with AS 1056, Storage water heaters; and
- should comply with AS/NZS 3500.4:2003, Plumbing and drainage – Heated waters services and AS 3500.4.1 – 1997, National Plumbing and Drainage – Hot water supply systems – Performance requirements.

Non-compliant systems may be refused connection or disconnected.

## **16. LV ToU Residential (N13r)**

This network tariff is for low voltage installations that are premises used wholly or principally as Private Residential Dwellings.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

Farm outbuildings may be connected on this tariff provided that the connection is through the meters for the farm residence.

An installation that is supplied under this tariff may be reassigned to network tariff N01 – General Network Residential, provided it remains a Private Residential Dwelling.

This network tariff may also be used in conjunction with the following additional network tariff:

- N06a – Controlled Energy.

A Type 6 meter is the minimum required for installations on this network tariff and must be capable of recording time of use data.

### **16.1. Use of system charges**

The use of system charges applicable for this network tariff are composed of the following charging components:

- (a) DUoS
  - (i) A fixed daily charge; and
  - (ii) An energy-based charge; the rate of the energy charge varies according to the time of day at which energy is consumed, with the periods being as identified in Table 2.
- (b) TUoS
  - (i) An energy-based charge; the rate of the energy charge varies according to the time of day at which energy is consumed, with the periods being as identified in Table 2.

### **16.2. Requirements of water heating systems**

In all installations, for connection on this network tariff, the water heating systems:

- must comply with AS 1056, Storage water heaters; and
- should comply with AS/NZS 3500.4:2003, Plumbing and drainage – Heated waters services and AS 3500.4.1 – 1997, National Plumbing and Drainage – Hot water supply systems – Performance requirements.

Non-compliant systems may be refused connection or disconnected.

**16.3. Time of use periods**

Table 2 sets out the time of use periods applicable to this network tariff.

**Table 2: Time periods for LV ToU - Residential**

| Time Period                                          | Tariff Rate |
|------------------------------------------------------|-------------|
| Week Day (07:00 – 11:00)<br>(Monday – Friday)        | Peak        |
| Week Day (11:00 – 16:30)<br>(Monday – Friday)        | Shoulder    |
| Week Day (16:30 – 22:00)<br>(Monday – Friday)        | Peak        |
| Weekend Day (07:00 – 22:00)<br>(Saturday and Sunday) | Shoulder    |
| Any Day (22:00 – 07:00)<br>(Monday – Sunday)         | Off-peak    |

## **17. LV ToU - Business (N13b/N13c)**

This network tariff is for low voltage installations that are not Private Residential Dwellings.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

A site that is supplied under this network tariff may not be reassigned to network tariffs N02a – General Network Business, Nursing Homes or N02b – General Network Business, Curtilage.

This network tariff may also be used in conjunction with the following additional network tariff:

- N06a – Controlled Energy.

A Type 6 meter is the minimum required for installations on this network tariff and must be capable of recording time of use data.

### **17.1. Use of system charges**

The use of system charges applicable for this network tariff are composed of the following charging components:

- (a) DUoS
  - (i) A fixed daily charge; and
  - (ii) An energy based charge; the rate of the energy charge varies according to the time of day at which energy is consumed, with the periods being identified in Table 3.
- (b) TUoS
  - (i) An energy-based charge; the rate of the energy charge varies according to the time of day at which energy is consumed, with the periods being as identified in Table 3.

### **17.2. Requirements of water heating systems**

In all installations, for connection on this network tariff, the water heating systems:

- must comply with AS 1056, Storage water heaters; and
- should comply with AS/NZS 3500.4:2003, Plumbing and drainage – Heated waters services and AS 3500.4.1 – 1997, National Plumbing and Drainage – Hot water supply systems – Performance requirements.

Non-compliant systems may be refused connection or disconnected.

### 17.3. Time of use periods

Table 3 sets out the time of use periods applicable to this network tariff.

**Table 3: Time periods for LV ToU - Business**

| Time Period                                          | Tariff Rate |
|------------------------------------------------------|-------------|
| Week Day (07:00 – 22:00)<br>(Monday – Friday)        | Peak        |
| Weekend Day (07:00 – 22:00)<br>(Saturday and Sunday) | Shoulder    |
| Any Day (22:00 – 07:00)<br>(Monday – Sunday)         | Off-peak    |

## **18. Small LV Unmetered (N07)**

This network tariff is for small, low voltage, low demand installations with a relatively constant load profile. For example:

- illuminated street signs;
- public telephone kiosks;
- electric fences;
- two-way radio transmitters;
- fixed steady wattage installations;
- traffic lights; or
- level crossings.

All installations on this network tariff must have all components permanently connected. For the avoidance of doubt, an installation containing a general purpose outlet does not qualify for this network tariff.

Exception: Shows and public functions qualify for this network tariff. Shows and public functions on this network tariff may have installed general purpose outlets.

This network tariff may not be used in conjunction with any other network tariff.

This is an unmetered network tariff and will be treated as a Type 7 metering installation.

## **19. LV Day/Night (Irrigation) (N08)**

This network tariff is obsolete and therefore no longer available to new customers.

This low voltage network tariff is for primary producer's business installations that are used solely for the irrigation of crops and must be classified as ANZSIC class 01.

This network tariff may not be used in conjunction with any other network tariff.

A Type 6 meter is the minimum required for installations on this network tariff and must have the ability to record "peak" and "off-peak" consumption.

This network tariff will be discontinued over a two year period and will not be available to any customer after 1 July 2014. Any customer currently connected to this network tariff may make application to Aurora for reassignment to another network tariff during the 2013-14 regulatory year. Aurora suggests that existing customers should seek reassignment to network tariff N08a/N08b – LV Irrigation (ToU).

Aurora will write to any customers remaining on this network tariff after 30 March 2013 advising that Aurora will reassign those customers to network tariff N08a/N08b – LV Irrigation (ToU) between 1 July 2013 and 30 June 2014.

### **19.1. Time of use availability**

This network tariff is a simple "time of use" tariff, with different rates for "peak" and "off-peak" consumption.

The off-peak rate will be available daily for not less than 10 hours between 20:00 hours and 07:00 hours the following day. In general, the "window" will be of 10 hours duration. Subject to network constraints and at the discretion of Aurora, an 11 hour window may be available upon request.

To provide a "staggered start" to ensure distribution network stability, the off-peak "window" for individual customers will start between 20:00 and 21:00, and end between 06:00 and 07:00 the following day.

Aurora will arrange the start and finish times.

## 20. LV Irrigation (ToU) (N08a/N08b)

This low voltage network tariff is for primary producer's business installations that are used solely for the irrigation of crops and must be classified as ANZSIC class 01.

This network tariff may not be used in conjunction with any other network tariff.

A Type 6 meter is the minimum required for installations on this network tariff and must be capable of recording time of use data.

### 20.1. use of system Charges

The use of system charges applicable for this network tariff are composed of the following charging components:

- (a) DUoS
  - (i) A fixed daily charge; and
  - (ii) An energy-based charge; the rate of the energy charge varies according to the time of day at which energy is consumed, with the periods being as identified in Table 4.
- (b) TUoS
  - (i) An energy-based charge; the rate of the energy charge varies according to the time of day at which energy is consumed, with the periods being as identified in Table 4.

### 20.2. Time of use periods

Table 4 sets out the time of use periods applicable to this network tariff.

**Table 4: Time periods for LV Irrigation (ToU)**

| Time Period                                          | Summer<br>(1 Oct – 31 Mar) | Winter<br>(1 Apr – 30 Sep) |
|------------------------------------------------------|----------------------------|----------------------------|
| Week Day (07:00 – 22:00)<br>(Monday – Friday)        | Shoulder                   | Peak                       |
| Weekend Day (07:00 – 22:00)<br>(Saturday and Sunday) | Off-peak                   | Shoulder                   |
| Any Day (22:00 – 07:00)<br>(Monday – Sunday)         | Off-peak                   | Off-peak                   |

## **21. LV kW Demand (N03)**

This network tariff is obsolete and therefore no longer available to new customers.

This network tariff is for installations taking low voltage 3-phase supply.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

This network tariff may not be used in conjunction with any other network tariff.

A Type 6 meter is the minimum required for installations on this network tariff.

This network tariff will be discontinued and will not be available to any customer after 1 July 2013. Any customer currently connected to this network tariff may make application to Aurora for reassignment to another network tariff during the 2012-13 regulatory year. Aurora suggests that existing customers should seek reassignment to network tariff N09 – LV kVA Demand.

Aurora will write to all customers remaining on this network tariff advising that Aurora will reassign those customers to network tariff N09 – LV kVA Demand prior to 30 June 2013.

## **22. LV kVA Demand (N09)**

This network tariff is for installations taking low voltage 3-phase supply.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

This network tariff may not be used in conjunction with any other network tariff.

A Type 6 meter is the minimum required for installations on this network tariff.

## **23. HV kVA Demand (N10)**

This network tariff is obsolete and therefore no longer available to new customers.

This network tariff is for customers taking supply at high voltage.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

The customer must supply its own transformers and switchgear for installations connected on this network tariff.

This network tariff may not be used in conjunction with any other network tariff.

Metering of consumption is at the HV connection point.

This network tariff will be discontinued and will not be available to any customer after 1 July 2013. Any customer currently connected to this network tariff may make application to Aurora for reassignment to another network tariff during the 2012-13 regulatory year. Aurora suggests that existing customers should seek reassignment to network tariff N10s – HV kVA Specified Demand.

Aurora will write to all customers remaining on this network tariff advising that Aurora will reassign those customers to network tariff N10s – HV kVA Specified Demand prior to 30 June 2013 and that the customer must set a specified demand with Aurora.

## **24. HV kVA Specified Demand (N10s)**

This network tariff is for installations taking supply at high voltage, with an expected ATMD less than 2 MVA.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

The customer must supply its own transformers and switchgear for installations connected on this network tariff.

This network tariff may not be used in conjunction with any other network tariff.

Metering of consumption for installations on this network tariff is at the HV connection point and must be capable of recording “interval data”.

### **24.1. Negotiation of specified demand**

Customers on this network tariff are able to agree with Aurora a specified demand for their electrical installation. Once agreed this value is used in the calculation of demand charges for the following period.

Negotiation of the specified demand for subsequent periods should be completed at least two months prior to the end of the last negotiated period.

Renegotiation of the specified demand is limited to one occurrence each 12 months or as otherwise agreed with Aurora.

### **24.2. Use of system charges**

The use of system charges applicable for this network tariff are composed of the following charging components:

- (a) DUoS
  - (i) A fixed daily charge; and
  - (ii) An energy-based charge; the rate of the energy charge varies according to the time of day at which energy is consumed, with the periods being as identified in Table 5; and
  - (iii) A demand-based charge calculated according to the method given in section 24.3. For the purposes of this calculation, the excess demand rate is 10 times the specified demand rate.
- (b) TUoS
  - (i) An energy-based charge; the rate of the energy charge varies according to the time of day at which energy is consumed, with the periods being as identified in Table 5; and
  - (ii) A demand-based charge calculated according to the method given in section 24.3. For the purposes of this calculation, the excess demand rate is 10 times the specified demand rate.

### **24.3. Calculation of demand charges**

The demand-based charges for an installation on this network tariff will be calculated monthly as follows:

- (a) the monthly demand-based charge is the sum of the daily demand-based charges for the month;

- (b) the daily demand-based charges are:
- (i) if any daily ATMD in the month is less than or equal to 100 per cent of the specified demand, the demand charge for the day will be equal to the specified demand multiplied by the specified demand rate; or
  - (ii) if the daily ATMD is greater than the specified demand, but not greater than 120 per cent of the Specified demand, then the demand charge for the day will be the maximum demand multiplied by the specified demand rate; or
  - (iii) if any daily ATMD in the month is greater than the specified demand by 120 per cent, the daily demand charge will be the sum of:
    - 120 per cent of the specified demand multiplied by the specified demand rate; and
    - the difference between the ATMD and 120 per cent of the specified demand multiplied by the excess demand rate.

#### 24.4. Time of use periods

Table 5 sets out the time of use periods applicable to this network tariff.

**Table 5: Time periods for HV kVA Specified Demand**

| Time Period                                          | Summer<br>(1 Oct – 31 Mar) | Winter<br>(1 Apr – 30 Sep) |
|------------------------------------------------------|----------------------------|----------------------------|
| Week Day (07:00 – 22:00)<br>(Monday – Friday)        | Shoulder                   | Peak                       |
| Weekend Day (07:00 – 22:00)<br>(Saturday and Sunday) | Off-peak                   | Shoulder                   |
| Any Day (22:00 – 07:00)<br>(Monday – Sunday)         | Off-peak                   | Off-peak                   |

## **25. HV kW Demand (N11)**

This network tariff is obsolete and therefore no longer available to new customers.

This network tariff is for customers taking supply at high voltage.

There are no restrictions on the use of the supply (i.e. the supply may be used for general power, heating, water heating, etc.).

The customer must supply its own transformers and switchgear for installations connected on this network tariff.

This network tariff may not be used in conjunction with any other network tariff.

Metering of consumption is at the HV connection point.

This network tariff will be discontinued and will not be available to any customer after 1 July 2013. Any customer currently connected to this network tariff may make application to Aurora for reassignment to another network tariff during the 2012-13 regulatory year. Aurora suggests that existing customers should seek reassignment to network tariff N10s – HV kVA Specified Demand.

Aurora will write to all customers remaining on this network tariff advising that Aurora will reassign those customers to network tariff N10s – HV kVA Specified Demand prior to 30 June 2013 and that the customer must set a specified demand with Aurora.

## **26. HV kVA Specified Demand (> 2.0 MVA) (N15)**

This network tariff applies to customers, with an ATMD in excess of 2.0 MVA, supplied directly from the Aurora distribution network with no Aurora owned assets beyond the connection point.

The customer must supply its own transformers and switchgear for HV installations connected on this network tariff.

A site connected to the Aurora distribution network with this network tariff is not eligible for any other network tariff.

Metering of consumption for installations on this network tariff is at the HV connection point and must be capable of recording “interval data”.

### **26.1. Negotiation of specified demand**

Customers on this network tariff are able to agree with Aurora a specified demand for their electrical installation. Once agreed this value is used in the calculation of demand charges for the following period.

Negotiation of the specified demand for subsequent periods should be completed at least two months prior to the end of the last negotiated period.

Renegotiation of specified demand is limited to one occurrence each twelve months or as otherwise agreed with Aurora.

### **26.2. Use of system Charges**

The use of system charges applicable for this tariff are composed of the following charging components.

- (a) DUoS
  - (i) A fixed daily charge; and
  - (ii) An energy-based charge; the rate of the energy charge varies according to the time of day at which energy is consumed, with the periods being as identified in Table 6; and
  - (iii) A demand-based charge calculated according to the method given in section 26.3. For the purposes of this calculation, the excess demand rate is 5 times the specified demand rate.
- (b) Connection
  - (i) A demand-based charge calculated according to the method given in section 26.3. For the purposes of this calculation, the excess demand rate is 5 times the specified demand rate.
- (c) TUoS
  - (i) A demand-based charge calculated according to the method given in section 26.3. For the purposes of this calculation, the excess demand rate is 5 times the specified demand rate.
  - (ii) The TUoS charges for customers connected on this network tariff are based on the actual charges received from the transmission network service provider (Transend) for the relevant transmission connection point. This provides the greatest cost reflectivity and preserves the pricing signals within the Transend charges for these customers.

### 26.3. Calculation of demand charges

The demand-based charges for an installation on the network tariff N15 will be calculated monthly as follows:

- (a) the monthly demand-based charge is the sum of the daily demand-based charges for the month;
- (b) the daily demand-based charges are:
  - (i) if any daily ATMD in the month is less than or equal to 100 per cent of the specified demand, the demand charge for the day will be equal to the specified demand multiplied by the specified demand rate; or
  - (ii) if any daily ATMD in the month is greater than the specified demand, the daily demand charge will be the sum of:
    - the specified demand multiplied by the specified demand rate; and
    - the difference between the ATMD and the specified demand multiplied by the excess demand rate.

### 26.4. Time of use periods

Table 6 sets out the time of use periods applicable to this network tariff.

**Table 6: Time periods for HV kVA Specified Demand (>2MVA)**

| Time Period                                          | Summer<br>(1 Oct – 31 Mar) | Winter<br>(1 Apr – 30 Sep) |
|------------------------------------------------------|----------------------------|----------------------------|
| Week Day (07:00 – 22:00)<br>(Monday – Friday)        | Shoulder                   | Peak                       |
| Weekend Day (07:00 – 22:00)<br>(Saturday and Sunday) | Off-peak                   | Shoulder                   |
| Any Day (22:00 – 07:00)<br>(Monday – Sunday)         | Off-peak                   | Off-peak                   |

## **27. Street Lighting (N20)**

This low voltage network tariff is for the provision of Aurora's public lighting services and is available to councils, road authorities and customers wishing to install contract lighting.

The street lighting tariff rate is based on a "use of system charge" and is charged on a per lamp wattage rate. This network tariff charge is an additional charge to that published by Aurora for the provision of public lighting services<sup>3</sup>.

This network tariff does not include charges for the installation and/or replacement of lamps. Costs for installation and/or replacement of lamps are an additional charge and are included in Aurora's public lighting services tariffs.

This network tariff may not be used in conjunction with any other network tariff.

This is an unmetered network tariff and will be treated as a Type 7 metering installation.

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<sup>3</sup> Aurora's public lighting services tariffs are discussed in Aurora's Public Lighting Services Application and Price Guide.

## **28. Import Energy (N21)**

This tariff is for those installations that import energy into the Distribution System.

A Type 6 meter is the minimum required for installations and a charge for the provision of basic metering services may apply.

## **29. Individual Tariff Calculation**

Individual Tariff Calculation (ITC) network prices will typically apply to customers with an electrical demand in excess of 2.0 MVA or where a customer's circumstances in a pricing zone identifies that the average shared network charge to be meaningless or distorted. Individually calculated customer network charges are determined by modelling the connection point requirements as requested by the customer or their agents.

ITC prices are based on actual Transmission use of system (TUoS) charges for the relevant transmission connection point (preserving the pricing signals within the Transend charges), plus charges associated with the actual shared distribution network utilised for the electricity supply, plus connection charges based on the actual connection assets utilised. This provides the greatest cost reflectivity for this type of customer and is feasible since the number of such customers is relatively small.

ITC pricing is also justified by virtue of the shared distribution network assets being designated specifically to meet the requirements of these customers. Where the designation of shared network assets is difficult due to the specific connectivity of the customer, Aurora will apply ITC pricing on a mutually agreed basis.

ITC pricing can also be influenced by the load factor of the Customer's installation.

ITC pricing for customers with electrical demand that is less than 2.0 MVA could occur where:

- a customer has a dedicated supply system that is different and separate from the remainder of the supply network;
- there are only a small number of customers in a supply system making average prices inappropriate; or
- inequitable treatment of otherwise comparable customers arises from the electrical demand lower limit of 2.0 MVA. Selection of these customers will be at Aurora's discretion.

### **30.        Embedded Generation**

Network tariff charges for embedded generation connections will always be treated on an individually calculated basis.

### 31. Locational TUoS charges

Locational TUoS charges for those customers supplied under network tariffs N15 – HV kVA Specified Demand (> 2.0 MVA) and ITC – Individual Tariff Calculation will apply for the transmission connection sites detailed in Table 7.

**Table 7: Transmission Connection Sites**

| <b>Transmission node description</b> | <b>Transmission node identifier</b> |
|--------------------------------------|-------------------------------------|
| Arthurs Lake                         | TAL2                                |
| Avoca                                | TAV2                                |
| Burnie                               | TBU3                                |
| Bridgewater                          | TBW2                                |
| Creek Road <sup>1</sup>              | TCR2                                |
| Chapel Street <sup>1</sup>           | TCS3                                |
| Derwent Bridge                       | TDB2                                |
| Derby                                | TDE2                                |
| Devonport                            | TDP2                                |
| Emu Bay                              | TEB2                                |
| Electrona                            | TEL2                                |
| George Town <sup>2</sup>             | TGT3                                |
| Hadspen <sup>2</sup>                 | THA3                                |
| Kerlandie                            | TKE2                                |
| Kingston                             | TKI2                                |
| Knights Road                         | TKR2                                |
| Lindisfarne <sup>1</sup>             | TLF2                                |
| Meadowbank                           | TMB2                                |
| Mornington <sup>1</sup>              | TMT2                                |
| Mowbray <sup>2</sup>                 | TMY2                                |
| North Hobart <sup>1</sup>            | TNH2                                |
| New Norfolk                          | TNN2                                |
| Newton                               | TNT2                                |
| Norwood <sup>2</sup>                 | TNW2                                |
| Port Latta                           | TPL2                                |
| Palmerston                           | TPM3                                |
| Queenstown                           | TQT2                                |
| Railton                              | TRA2                                |
| Rosebery                             | TRB2                                |

| Transmission node description | Transmission node identifier |
|-------------------------------|------------------------------|
| Risdon <sup>1</sup>           | TRI4                         |
| Rokeby <sup>1</sup>           | TRK2                         |
| Scottsdale                    | TSD2                         |
| St Leonards <sup>2</sup>      | TSL2                         |
| St Marys                      | TSM2                         |
| Sorell                        | TSO2                         |
| Savage River                  | TSR2                         |
| Smithton                      | TST2                         |
| Triabunna                     | TTB2                         |
| Trevallyn <sup>2</sup>        | TTR2                         |
| Tungatinah                    | TTU2                         |
| Ulverstone                    | TUL2                         |
| Waddamana                     | TWA2                         |
| Hobart Virtual                | TVN1                         |
| Tamar Virtual                 | TVN2                         |
| Wesley Vale                   | TWV2                         |
| Wayatinah                     | TWY2                         |

1 Charges for these transmission nodes are averaged as discussed below.

2 Charges for these transmission nodes are averaged as discussed below.

### 31.1. Virtual nodes

Due to the interconnected nature of the Hobart region, transmission nodes (TCS3, TCR2, TLF2, TMT2, TNH2, TRI4 and TRK2) are averaged as a single Virtual Transmission Node (VTN) in accordance with the provisions of the Rules. The transmission node identifier for this VTN is TVN1.

**Table 8: Hobart region virtual transmission nodes**

| Transmission node identifier | Transmission node description |
|------------------------------|-------------------------------|
| TCR2                         | Creek Road                    |
| TCS3                         | Chapel Street                 |
| TLF2                         | Lindisfarne                   |
| TMT2                         | Mornington                    |
| TNH2                         | North Hobart                  |
| TRI4                         | Risdon                        |
| TRK2                         | Rokeby                        |

Due to the interconnected nature of the Launceston/Tamar region, transmission nodes (TGT3, THA3, TMY2, TNW2, TSL2 and TTR2) are averaged as a single VTN in accordance with the provisions of the Rules. The transmission node identifier for this VTN is TVN2.

**Table 9: Tamar region virtual transmission nodes**

| <b>Transmission node identifier</b> | <b>Transmission node description</b> |
|-------------------------------------|--------------------------------------|
| TGT3                                | George Town                          |
| THA3                                | Hadspen                              |
| TMY2                                | Mowbray                              |
| TNW2                                | Norwood                              |
| TSL2                                | St Leonards                          |
| TTR2                                | Trevallyn                            |

## 32. Definitions

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AER                                   | Australian Economic Regulator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ATMD                                  | The maximum demand at anytime during a day within a billing period.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| AS/NZS                                | Australia and New Zealand Standards.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Aurora                                | Unless otherwise stated means Aurora Energy Pty Ltd ABN 85 082 464 622 in its capacity as a Distribution Network Service Provider.                                                                                                                                                                                                                                                                                                                                                                                 |
| Aurora Energy Business                | Unless otherwise stated means Aurora Energy Pty Ltd ABN 85 082 464 622 in its capacity as a retailer.                                                                                                                                                                                                                                                                                                                                                                                                              |
| billing period                        | The period covered by the bill sent to a retailer or customer.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Connection point                      | In relation to a Customer, the point at which electricity leaves the Distribution System for delivery to the Customer provided that where the Customer's Electrical Installation is not directly connected to the Distribution System, the Connection Point is the point at which the electricity last leaves the Distribution System before being delivered to the Customer, whether or not the electricity passes through facilities owned or operated by another person before being delivered to the Customer. |
| Contestable Customer Regulations      | Electricity Supply Industry (Contestable Customer) Regulations 2005.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Customer                              | As defined in the Tasmanian Electricity Code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Distribution network                  | As defined in the Tasmanian Electricity Code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Distribution network service provider | A person who engages in the activity of owning, controlling, or operating a Distribution System.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Distribution system                   | As defined in the NER.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Distribution use of system (DUoS)     | A charge to a Distribution Network user for use of the Distribution System for the conveyance of electricity.                                                                                                                                                                                                                                                                                                                                                                                                      |
| Embedded generation                   | A generation unit connected within a Distribution System and not having direct access to a Transmission System.                                                                                                                                                                                                                                                                                                                                                                                                    |
| Electrical contractor                 | An electrical contractor is a business person who is highly trained and is also registered to employ electricians to help install and design electricity.                                                                                                                                                                                                                                                                                                                                                          |
| Excess demand                         | The difference between the Specified Demand and the Maximum Demand – the additional demand for a specific period and charged at the agreed charge rate.                                                                                                                                                                                                                                                                                                                                                            |
| Generation unit                       | The actual generator of electricity and all the related equipment essential to its functioning as a single entity.                                                                                                                                                                                                                                                                                                                                                                                                 |
| high voltage (HV)                     | A voltage exceeding 1,000 volts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Interval metering services            | Reading services for interval meters – types 1-5 as defined in the NER and provision of other services.                                                                                                                                                                                                                                                                                                                                                                                                            |
| Irrigation                            | The provision of water pumping capability that facilitates primary production.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ITC                                   | Has the meaning “Individual Tariff Calculation”.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Load factor                           | This is the ratio of a Customer's average electrical load divided by the maximum electrical load.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| low voltage (LV)                      | A voltage not exceeding 1,000 volts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum demand                       | The highest amount of electrical power delivered, or forecast to be delivered, over a defined period (day, week, month, season or year) either at a connection point, or simultaneously at a defined set of connection points.                                                                                                                                                                                                                                                                                          |
| National electricity market (or NEM) | As defined in the Tasmanian Electricity Code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| NER or Rules                         | National Electricity Rules.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Network                              | The apparatus, equipment, plant and buildings used to convey, and control the conveyance of, electricity to customers (whether wholesale or retail) excluding any connection assets. In relation to a Network Service Provider, a network owned, operated or controlled by that Network Service Provider.                                                                                                                                                                                                               |
| Network tariff                       | Means the schedule of fees (including the rate or rates) Aurora uses to calculate the amount it charges customers, or a class of customers, for network services, and for network services and metering services associated with a Type 5, 6 or 7 installation, as amended from time to time.                                                                                                                                                                                                                           |
| Obsolete tariff                      | Network tariffs that have been superseded but remain in place until such time as they are rescinded or the electrical configuration of a Customer's installation is altered.                                                                                                                                                                                                                                                                                                                                            |
| Published tariffs                    | Those network tariffs published from time to time, usually annually, by Aurora.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Private Residential Dwelling         | A reference to a private residential dwelling is a house, unit, town house or apartment and is a reference to an establishment that, in the reasonable opinion of Aurora is not classifiable under the Australian and New Zealand Standard Industrial Classification (ANZSIC). The ANZSIC system is based on businesses, and it is used to classify businesses. It includes any organisation, which provides goods and services, including companies, non-profit organisations, government departments and enterprises. |
| Registered Electrician               | A Person or Company licensed under the Electricity Industry Safety and Administration Act 1997 and the Occupational Licensing Act 2005 to perform maintenance, alteration or installation work on electrical infrastructure and associated fittings.                                                                                                                                                                                                                                                                    |
| Retailer of choice                   | The customer's current or chosen electricity retailer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Specified demand                     | Means the value of the electrical demand at the site to which the Specified Demand network tariff applies as nominated by the operator of that site to Aurora.                                                                                                                                                                                                                                                                                                                                                          |
| Supply voltage                       | The nominal voltage measured at the Connection Point.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Time of use                          | A tariff application that has variable rates dependent on the time of day electricity is consumed.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Transend                             | Unless otherwise stated means Transend Networks Pty Ltd ABN 57 082 586 892 in its capacity as a Transmission Network Service Provider.                                                                                                                                                                                                                                                                                                                                                                                  |
| Transmission network                 | As defined in the Tasmanian Electricity Code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Transmission system                  | As defined in the Tasmanian Electricity Code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Transmission use of system (TUoS)    | A charge to a Transmission Network user for use of the Transmission System for the conveyance of electricity.                                                                                                                                                                                                                                                                                                                                                                                                           |
| Use of system (UoS)                  | A charge to a Network user for use of the Distribution and or Transmission System for the conveyance of electricity.                                                                                                                                                                                                                                                                                                                                                                                                    |